

Database Systems		
Lab #:	03	
Lab Title:	Designing of Forms, Reports creation of Relationships.	
Submitted by:		
Names		Registration #
Sumaira Safeer		FA22-BCE-019

Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab No: 03

Title: Designing of Forms, Reports creation of Relationships.

Objective:

The objective of this lab is to learn how to design a forms and reports, how to create relationships between tables and how to design a query. Forms provide a nice interface for entering data into tables and for viewing existing data. Reports provide you with a read-only, printable view of tables, queries or a combination of them. Tables are related by relationships in order to minimize data redundancy. Such redundancy is both wasteful and can lead to anomalies as explained in your text.

Types of table relationships:

- A one-to-many relationship:
- A many-to-many relationship:
- A one-to-one relationship

Task 01:

Fill in the blank record then use the arrow buttons to navigate between your records. Make sure you save your form. You will also notice that the same validation rules apply in the form.

Design View:

Employees	
Field Name	Data Type
SIN	Memo
FName	Text
LName	Text
TelNum	Memo
Birthdate	Date/Time
Date joined	Date/Time
Salary	Currency
Address	Text

City	Text
Postal Code	Number
On leave	Yes/No
Gender	Text

Datasheet View:

Employees							
SIN	FName	LName	TelNum	Birthdate	Date joined	Salary	
987-654-321	jane	Smith	(403)123-4567	2/8/1989	5/8/2009	\$65,000	
123-456-789	john	doe	() 123-4567	3/8/1986	7/13/2010	\$60,000	
*							

Address	City	Postal Code	On leave	Gender
Alberta, Canada	Calagary	4567	☐	Male
Alberta, canada	Calagary	5678	☒	Male

Form:

SIN	123-456-789	Date joined	7/13/2010
FName	john	Salary	\$60,000.00
LName	doe	Address	Alberta, canada
TelNum	() 123-4567	City	Calagary
Birthdate	3/8/1986	Postal Code	5678
		On leave	☒
		Gender	Male

Employees

SIN

987-654-321

^

v

Date joined

5/8/2009

FName

jane

Salary

\$65,000.00

LName

Smith

Address

Alberta, Canada

TelNum

(403)123-4567

City

Calagary

Birthdate

2/8/1989

Postal Code

4567

On leave

☐

Gender

Male

Report:

Employees					Saturday, October 12, 2024	
					3:43:30 PM	
SIN	FName	LName	TelNum	Birthdate		
123-456-789	john	doe	() 123-4567	3/8/1986		
987-654-321	jane	Smith	(403)123-4567	2/8/1989		

Date joined	Salary	Address	City	Postal Code	On leave	Gender
7/13/2010	\$60,000.00	Alberta, canada	Calagary	5678	☒	Male
5/8/2009	\$65,000.00	Alberta, Canada	Calagary	4567	☐	Male

\$125,000.00

Page 1 of 1

Report Wizard:

Employess at leave

SIN	FName	TelNum	
123-456-789	john	() 123-4567	☒
987-654-321	jane	(403)123-4567	☐

Task 02:

Design a relationship between Employee and department table, as in one department more than one employee can work.

Design View:

The screenshot shows the Microsoft Access Design View for the 'Department' table. The left pane shows the 'All Access Objects' list with 'Tables' selected, showing 'Department', 'Employee', and 'Query1'. The main area displays the table structure with the following fields:

Field Name	Data Type
ID	AutoNumber
Name	Short Text
DEPARTMENT	Short Text
University	Short Text
City	Short Text

Below the field list, the 'General' tab is active, showing various field properties:

Property	Value
Field Size	255
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	Yes
Allow Zero Length	Yes
Indexed	Yes (No Duplicates)
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

At the bottom, a status bar indicates: 'Design view. F6 = Switch panes. F1 = Help.'

Datasheet View:

The screenshot shows the Microsoft Access Datasheet View for the 'Department' table. The table contains the following data:

ID	Name	DEPARTMENT	University	City	Click to Add
2	Ali Mustafa	Department of Mathematics	Comsats University Islamabad Attock	Attock	
4	Arham Niaz	Department Of Managment sciences	Qaid-e-Azam University	Islambad	
3	Babar Ali	Department of Electrical & Computer Engineeering	University of Mianwali	Mianwali	
1	Farhan Khan	Department of Electrical & Computer Engineeering	University of Wah	Wah	
6	Harron	Department of Computer Science	Suzo Future University	Rawalpindi	
7	Quddoss Awan	Department of Mathematics	University of Sargodah	Sargodah	
8	Sharaz Khan	Department Of Managment sciences	Fatima Jinnah University	Faislabad	
5	Zeeshan Malik	Department of Computer Science	Lums University Lahore	Lahore	
*	(New)				

Design View:

Task 03:

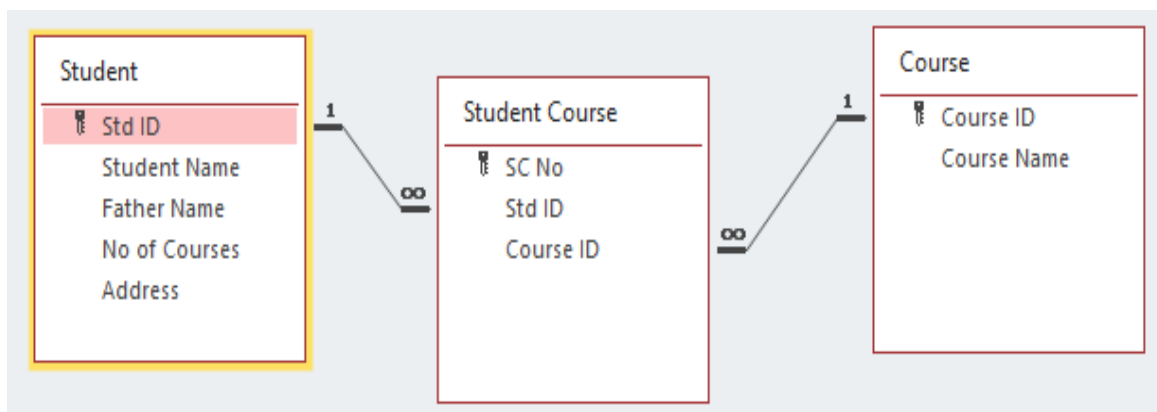
Create a many to many relationships between student and course as 1 student is reading more than one course and one course is teach to more than 1 students.

Datasheet View:

Relationships		Course	Student	Student Course	
Std ID	Student Name	Father Name	No of Courses	Address	
801	Hassaan	Shoukat	6	Attock	
SC No		Course ID	Click to Add		
E1		CSC 322			
C2		EEE 548			
*					
809	Zain	Muhammad	5	Fateh Jang	
*			0		

Relationships	Course	Student
Course ID	Course Name	Click to Add
CSC 322	Operating System	
CSC 371	Database System	

Relationship:



Post lab Task:

Postlab Task 01:

Design a form (Name Student Info) using the data given below.

- Apply the background image (Comsats) to the form and a Comsats logo in the header.
- Apply formatting to the form.
- Add buttons in the form (Add record, delete record, print form, go to first record, go to next record).
- Click on the button Add record and add a new record in the table.
- Check if the new record is added to the table.

id	student_regno	student_name	academic_session	academic_term	class_name	class_abbrev	class_group	subject_name	student_mark	category
1	OIS20160006	Ubah Obinna Tony	2016/2017	First Term	Junior Secondary School 1	JSS 1	A	English Language	7	Assignment
2	OIS20160006	Ubah Obinna Tony	2016/2017	First Term	Junior Secondary School 1	JSS 1	A	English Language	16	Test
3	OIS20160006	Ubah Obinna Tony	2016/2017	First Term	Junior Secondary School 1	JSS 1	A	Agricultural Science	9	Assignment
4	OIS20160006	Ubah Obinna Tony	2016/2017	First Term	Junior Secondary School 1	JSS 1	A	Agricultural Science	13	Test

All Acc...
Search...
Tables
NAME STUDENT I...
Forms
NAME STUDENT I...

NAME STUDENT INFO

Field Name	Data Type
ID	AutoNumber
Student Info	Short Text
Name	Short Text
Session	Short Text
Term	Short Text
Department	Short Text
Group	Short Text
Course	Short Text
Marks	Short Text
Category	Short Text

General
Lookup

Field Size	255
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	No
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

Datasheet View:

NAME STUDENT INFO										
ID	Student Info	Name	Session	Term	Department	Group	Course	Marks	Category	Click
1 56898		Hamza Ali	2022-2026	First	SE	A	Database	8	Assignment	
2 67768		Sumaira	2021-2025	Second	CE	B	Operating System	7	Quiz	
3 79907		Saqib Khan	2022-2026	First	BBA	A	OOP	9	Assignment	
4 57989		Zoya Naiq	2020-2024	Third	CS	C	Linear Algebra	10	Quiz	
5 56676		Tayyaba	2021-2025	Second	EE	B	Calculus	8	Quiz	
6 56678		Farhan Khan	2023-2027	First	BEN	A	Signal System	6	Assignment	
7 56678		Nazish Malik	2022-2026	First	SE	B	PF	8	Quiz	
* (New)										

Forms:

All Acc...
Search...
Tables
NAME STUDENT I...
Forms
NAME STUDENT I...

NAME STUDENT INFO
NAME STUDENT INFO

ID
1
Department
SE

student info
56898
group
A

name
Hamza Ali
Course
Database

session
2022-2026
Marks
8

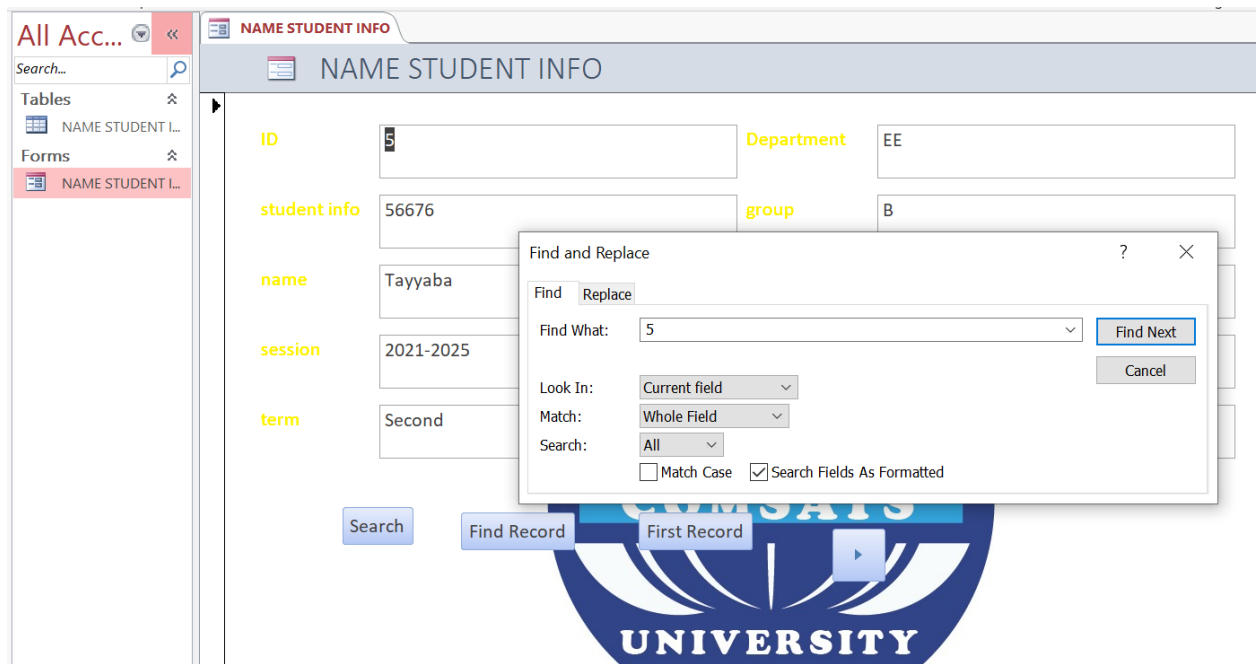
term
First
category
Assignment

Search
Find Record
First Record

COMSATS UNIVERSITY

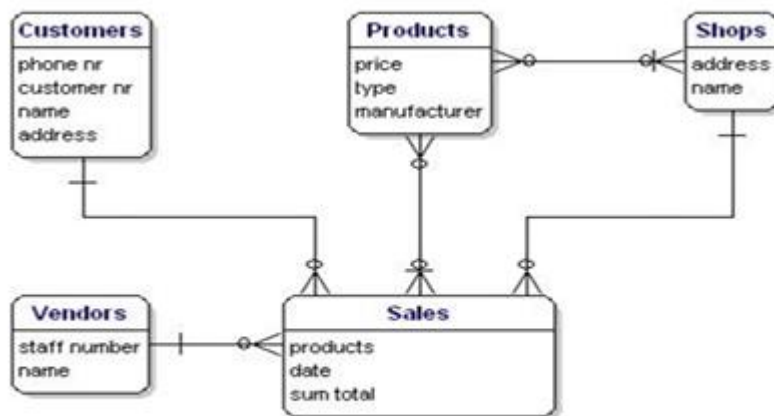
Record: 1 of 7
No Filter
Search

Finding Records:



Postlab Task 02:

Design these relationships after creating the tables.



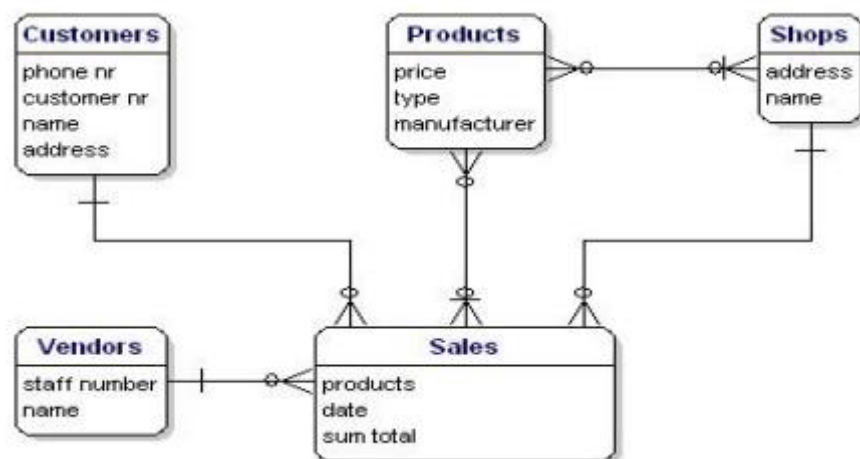
Customers - Sales: 1 customer can buy something several times.

- Sales - Customers: 1 sale is always made by 1 customer at the time.

- Vendors - sale: 1 vendor can do multiple sales.
- Sales - vendor: 1 sale can be done by only 1 vendor at a time.
- Shops - Products: in 1 shop there are multiple products.
- Products - Shops: 1 product (type) can be sold in multiple shops.
- Shops - Sales: in 1 shop multiple sales can be made.
- Sales - Shops: 1 sale can only be made in 1 shop at the time.
- Products - Sales: 1 product (type) can be purchased in multiple sales.
- Sales - Products: 1 sale can exist out of multiple products.

Relationships:

- Customers - Sales: 1: N
- Sales - Products: M: N
- Shops - Sales: 1: N
- Shops - Products: M: N
- Vendors - sales: 1: N
- Design these relationships after creating the tables.



- Customers → Sales: 1 customer can buy something several times.
- Sales → Customers: 1 sale is always made by 1 customer at the time.
- Vendors → sale: 1 vendor can do multiple sales.
- Sales → vendor: 1 sale can be done by only 1 vendor at a time.
- Shops → Products: in 1 shop there are multiple products.
- Products → Shops: 1 product (type) can be sold in multiple shops.
- Shops → Sales: in 1 shop multiple sales can be made.
- Sales → Shops: 1 sale can only be made in 1 shop at the time.
- Products → Sales: 1 product (type) can be purchased in multiple sales.
- Sales → Products: 1 sale can exist out of multiple products.

Relationships:

- Customers → Sales: 1: N
- Sales → Products: M: N
- Shops → Sales: 1: N
- Shops → Products: M: N
- Vendors → sales: 1: N

Datasheet View:

Customer No	Customer Name	Contact Number	Address
RDF4785	Fakhar	(333) 123-4568	Attock
SXD147	Khushdil	(300) 147-8523	Lahore

Sale No	Sale Date	Customer N	Staff No	Shop No	Sum Total
RF 123	20/03/2021	SXD147	568	RF 123	Rs. 5,000
YUT 753	31/08/2022	RDF4785	741	EE 753	Rs. 1,800

Product Code	Sales No	Quantity
AJK 7458	YUT 753	156
FDJ 1469	FRD 456	780

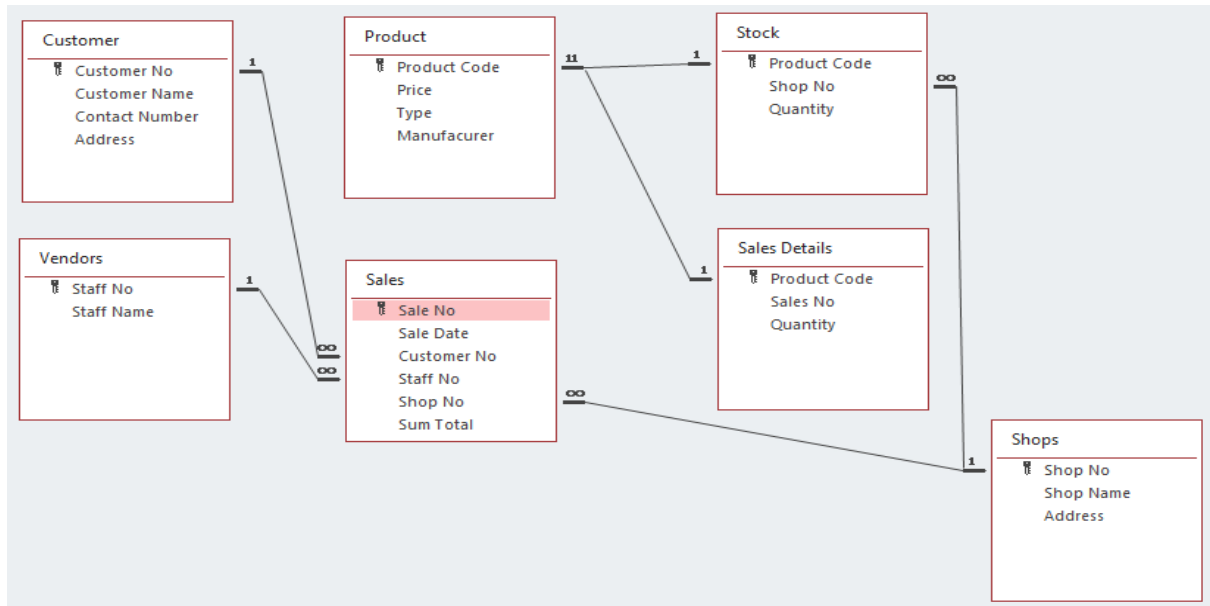
Shop No	Shop Name	Address	Click to Add
EE 753	TK Traders	Virtual	
RF 123	Islam Store	Attock	

Staff No	Staff Name
568	Ahmed
741	Ali

Product Code	Price	Type	Manufacturer
AJK 7458	Rs. 550	Electronics	Audionic
FDJ 1469	Rs. 1,000	Kitchen	Shan

Product Code	Shop No	Quantity
AJK 7458	EE 753	10
FDJ 1469	RF 123	60

Relationship:



Conclusion:

From this lab, we conclude that the tables are interconnected to each other. They can be shown in form of *Forms* and *Reports*. The relationship between them can also be shown.